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标题	Cascade aza-Wittig/6 π -Electrocyclization in the Synthesis of 1,6-Dihydropyridines
作者	Polychronidou V., Krupp A., Strohmann C., Antonchick A.
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摘要（原文节选）

A metal-free protocol for the synthesis of substituted 1,6-dihydropyridines with quaternary stereogenic centers via a cascade aza-Wittig/6 π -electrocyclization process has been developed. The high functional group compatibility and broad scope of this method were demonstrated by using a wide range of easily available vinyliminophosphoranes and ketones, with yields up to 97%. A modification of the obtained products allowed for an increase in complexity and chemical diversity. Finally, attempts for asymmetric synthesi ⋯

数据表 · Table 1（8 行，含数值行 7 行）

Optimization of Reaction Conditionsa

					yield (%)b	
entry	PR3	solvent [0.1 M]	temp (° C)	time (h)	3a	4a
1	PPh3	CH2Cl2	rt	72	20	–
2	PPh2Me	CH2Cl2	rt	48	70	–
3	PMe3	CH2Cl2	rt	12	92	–
4	PMe3	CH2Cl2	rt	96	92	–
5	PPh3	CHCl3	60	72	25	30
6	PMe3	CHCl3	60	72	–	84
7	PMe3	PhMe	110	72	–	80

脚注（逐字）：aReaction conditions: 1a (0.15 mmol) and 2a (0.15 mmol) were stirred at given temperature for given time.bIsolated yield.

数据表 · Table 2（9 行，含数值行 9 行）

Preliminary Investigation of the Asymmetric Electrocyclization of Substrate 3ba

entry	cat.	temp (° C)	yield (%)b	ee (%)c
1	(R)-7a	60	90	5
2	(S)-7b	60	88	3
3	(R)-7c	60	90	16
4	(R)-7d	60	70	13
5	(R)-7e	60	75	24
6	(R)-7e	50	40	34
7	(R)-7e	30	traces	–
8	8	60	80	3
9	9	60	85	2

脚注（逐字）：aReaction conditions: 3b (0.025 mmol) in CHCl3 [0.1 M] using 20 mol % of chiral catalyst for 72 h.bIsolated yield.cValues of ee were determined using chiral HPLC.

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